

Lecture 4 – Parameter Estimation & Hypothesis Testing

INT104 Artificial Intelligence

Parameter Estimation & Hypothesis Testing

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Concepts

- Point Estimate
- Confidence Level



The confidence level is the probability $1 - \alpha$ (such as 0.95, or 95%) that the confidence interval actually does contain the population parameter, assuming that the estimation process is repeated a large number of times.

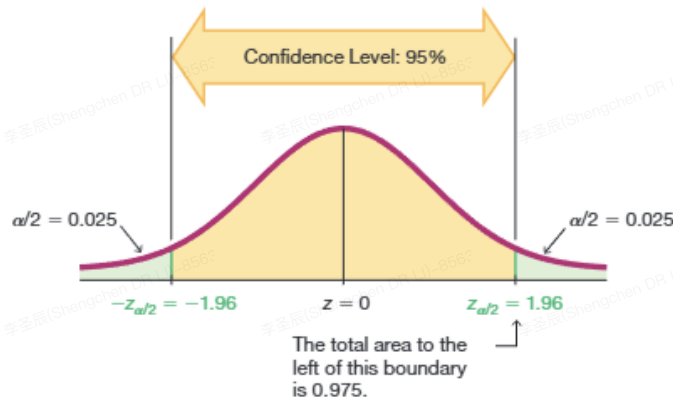
- Critical Value



For the standard normal distribution, a critical value is a z score on the borderline separating those z scores that are significantly low or significantly high.

Confidence Interval

This is our confidence level:	The area in both tails is:	The area in the right tail is:	The cumulative area from the left, excluding the right tail, is:
95%	$\rightarrow \alpha = 0.05$	$\rightarrow \alpha/2 = 0.025$	$\rightarrow 1 - 0.025 = 0.975$

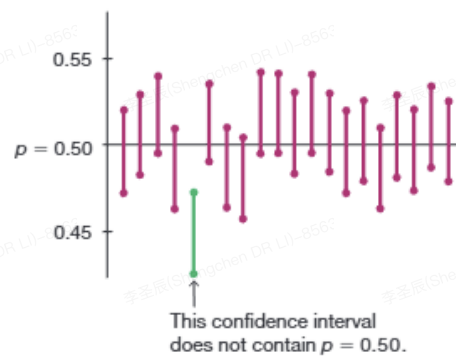


Confidence Interval

✓ We are 95% confident that the interval from 0.499 to 0.562 actually does contain the true value of the population proportion p .

✗ There is a 95% chance that the true value of p will fall between 0.499 and 0.562.

✗ 95% of sample proportions will fall between 0.499 and 0.562.



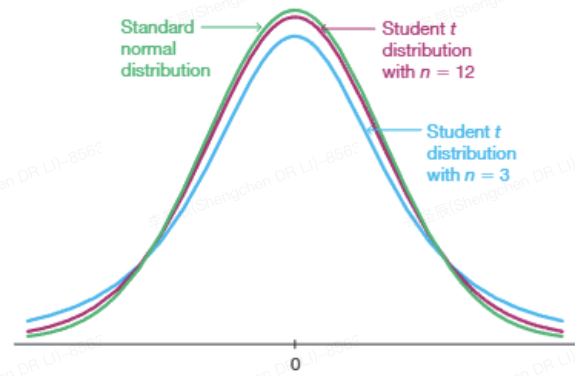
Determining Sample Size

- Z score separating an area of $\alpha/2$ in the right tail of the standard normal distribution
- Sample proportion (if known)
- Desired margin of error

Estimating a Population Mean

- The sample mean $\bar{x}$ is the best point estimate of the population mean μ .
 - For $n > 30$, assume normal distribution
 - For $n < 30$, assume student t distribution

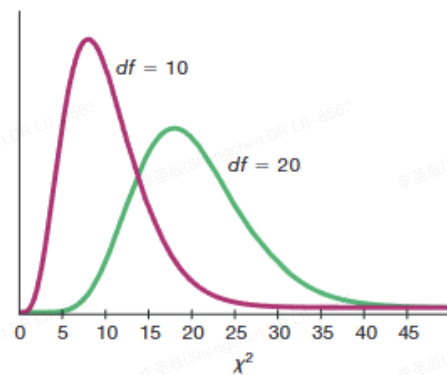
$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$



Estimate a Population Variance

Chi-Square Distribution

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2}$$



Hypothesis



In statistics, a hypothesis is a claim or statement about a property of a population.

A hypothesis test (or test of significance) is a procedure for testing a claim about a property of a population.

- A is larger than B
- A equals to [a constant]

Null and Alternative Hypothesis

- Reject null hypothesis H_0 to enable the alternative hypothesis H_a

Scenario: We are testing whether a new fertilizer (Fertilizer B) affects plant growth compared to a standard fertilizer (Fertilizer A).

- **Null Hypothesis (H_0):** There is no difference in average plant growth between plants treated with Fertilizer A and plants treated with Fertilizer B.
 - *Formally:* $H_0 : \mu_A = \mu_B$ (The mean growth for Group A equals the mean growth for Group B)
- **Alternative Hypothesis (H_1 or H_a):** There *is* a difference in average plant growth between plants treated with Fertilizer A and plants treated with Fertilizer B.
 - *Formally:* $H_a : \mu_A \neq \mu_B$ (The mean growth for Group A does *not* equal the mean growth for Group B)

Significance Level α

- The significance level α for a hypothesis test is the probability value used as the cutoff for determining when the sample evidence constitutes significant evidence against the null hypothesis
- By its nature, the significance level α is the probability of mistakenly rejecting the null hypothesis when it is true.



significance level $\alpha = P(\text{rejecting } H_0 \text{ when } H_0 \text{ is true})$

Determine the Parameter to Test

- In most cases, the mean will be tested for comparison
- However, it is also possible to examine variance (standard deviation) and proportion
- Choosing different parameters lead to different sampling distribution in the test



Mean $\rightarrow$ Student

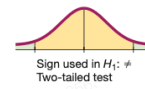
Variance $\rightarrow$ Chi-Square Distribution

Proportion $\rightarrow$ Normal Distribution

Define the Type of Test

- Type of Tests

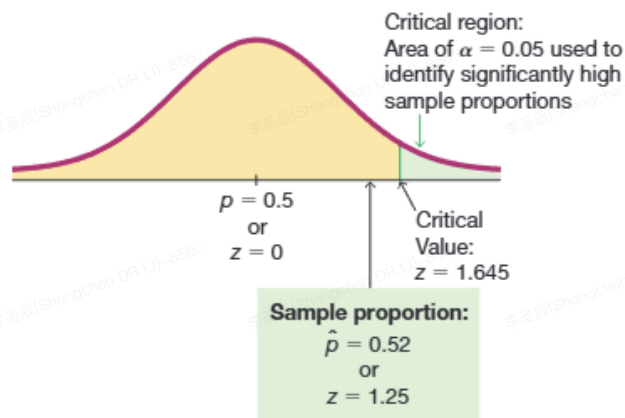
- Two-tailed test ($\neq$)
- Left-tailed test ($<$)
- Right-tailed test ($>$)



The different types of test lead to different z score selected

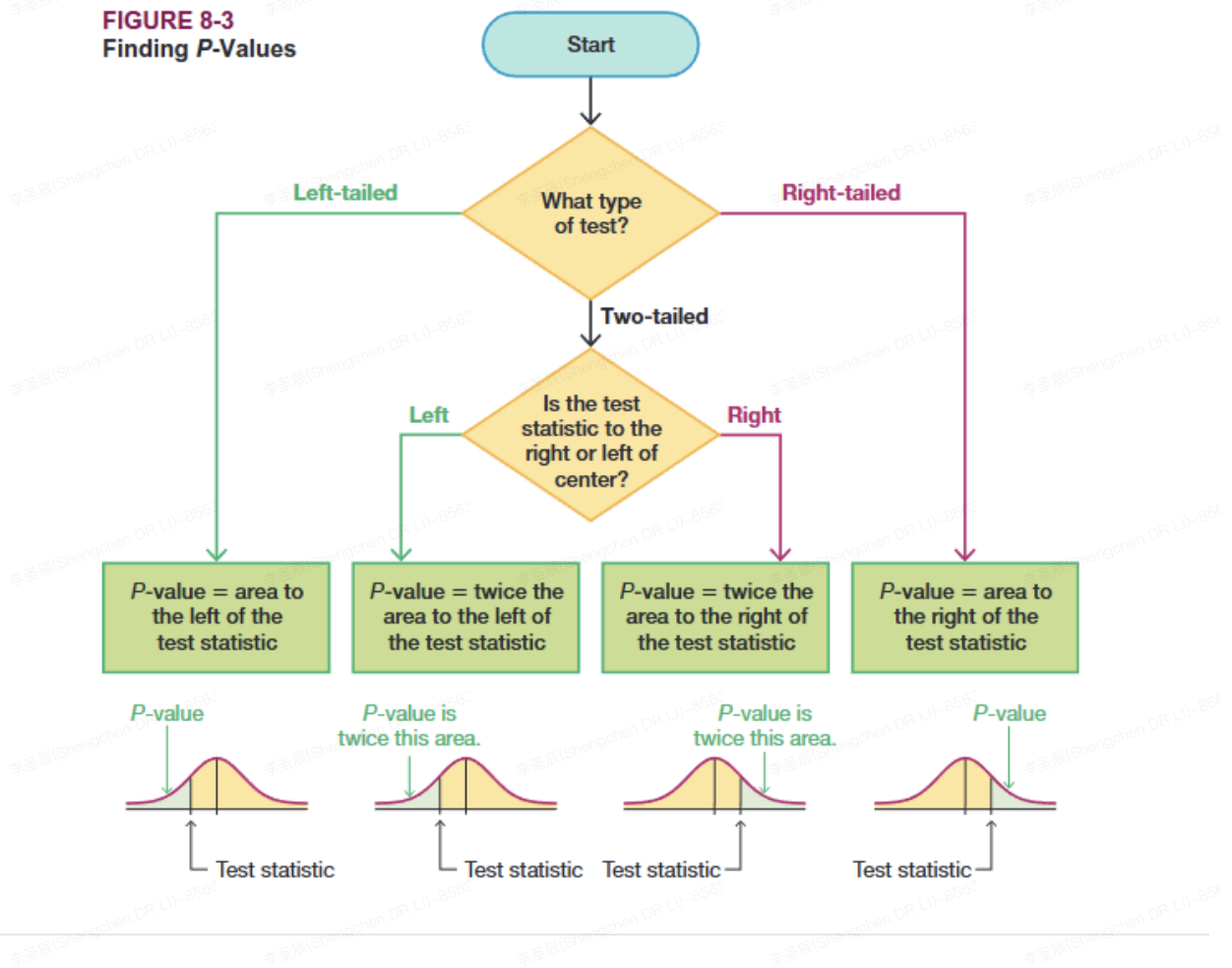
Select P-Value

- The selection of p value leads to different critical value



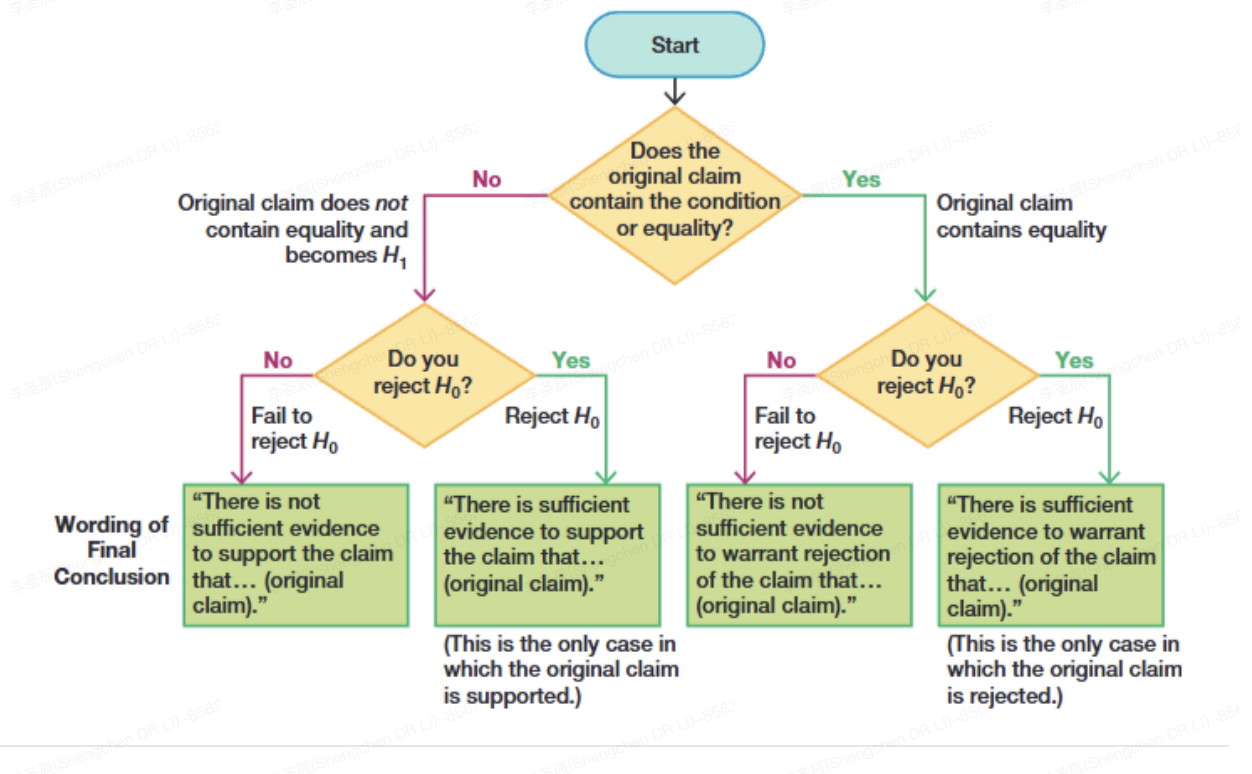
Select P-Value

FIGURE 8-3
Finding P-Values



Decision Making

- If $P\text{-value} \leq \alpha$, Reject $H_0 \rightarrow H_a$ Upheld.



Two Samples



We used hypothesis test to examine the value of a variable. What if we want to compare two variables?

Essentially, we have

$$H_0 : p_1 = p_2$$

$$H_1 : p_1 \neq p_2$$

We actually need to test

$$H_0 : p_1 - p_2 = 0$$

$$H_a : p_1 - p_2 \neq 0$$



What if we want to demonstrate $H_1 : p_1 > p_2$?
What assumption we need to make?

ANOVA



One-way analysis of variance (ANOVA) is a method of testing the equality of three or more population means by analyzing sample variances.

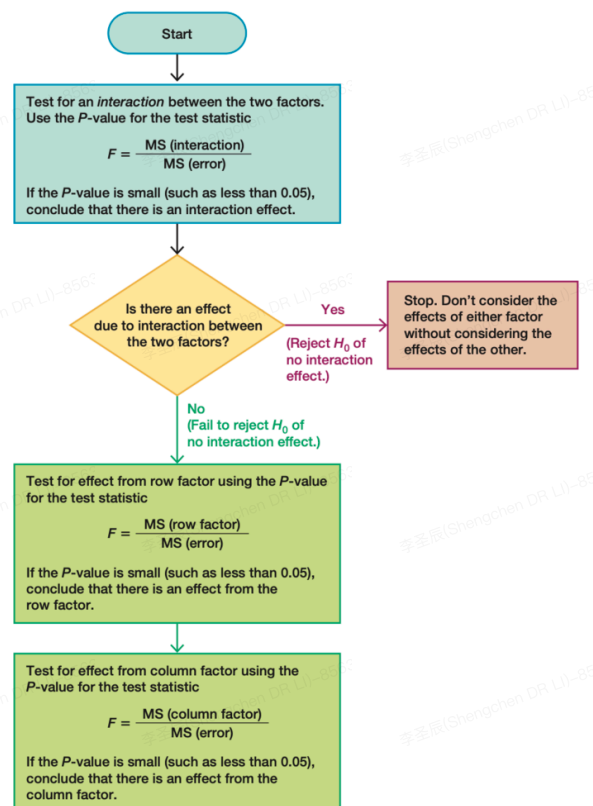
Requirements:

- The populations have distributions that are approximately normal.
- The populations have the same variance σ^2 (or standard deviation σ).
- The samples are simple random samples of quantitative data.
- The samples are independent of each other.
- The different samples are from populations that are categorized in only one way.

Two-way ANOVA

- With sample data categorized with two factors (a row variable and a column variable), use two-way analysis of variance to conduct the following three tests:

- Test for an effect from an interaction between the row factor and the column factor.
- Test for an effect from the row factor.
- Test for an effect from the column factor.



Non-parametric Tests

- Nonparametric (or distribution-free) tests do not require that samples come from populations with normal distributions or any other particular distributions.

Advantages

- Because nonparametric tests have less rigid requirements than parametric tests, they can be applied to a wider variety of situations.
- Nonparametric tests can be applied to more data types than parametric tests (particularly useful for ranks).

Disadvantages

- Nonparametric tests tend to waste information because exact numerical data are often reduced to a qualitative form.
- Nonparametric tests are not as efficient as parametric tests, so a nonparametric test generally needs stronger evidence (such as a larger sample or greater differences) in order to reject a null hypothesis.